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Education and the Decline in Average Hours Worked

Evidence from Nordic Labor Markets

Mauricio Vargas

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Education and the Decline in Average Hours Worked: Evidence from Nordic Labor Markets

Prepared by Mauricio Vargas*

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October 2026

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ABSTRACT: Average actual hours worked (AAHW) have declined across many advanced economies, affecting aggregate labor input and potential output. This paper examines the evolution of AAHW in Nordic countries between 2011 and 2022, with a focus on educational attainment and complementary ++gender, employment status, and citizenship. Using microdata from the European Union Labour Force Survey and a Shapley decomposition, the analysis shows that the decline in AAHW is primarily driven by reductions in hours among workers with lower and medium levels of educational attainment, used here as a proxy for skill levels. By contrast, highly educated workers work longer hours and their increasing share partly offsets the decline. Our decompositions show that this education-based result remains robust when worker groups are further split by age, gender and full-time/part-time status, and that most of the citizenship-related variation is concentrated among citizens, with non-citizen contributions generally smaller and often offsetting. Finland represents a temporary pandemic and post-pandemic exception, although the overall contribution of tertiary-educated workers remains positive. Supplementary evidence comparing desired and usual hours shows larger positive gaps among less-educated workers, while tertiary-educated workers in several countries appear closer to, or at times above, their preferred hours. These patterns are consistent with constraints on realized hours, stronger worker demand for hours, or both, but do not identify the underlying mechanism.

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WORKING PAPERS

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Glossary

AAHW	Average Actual Hours Worked in Main Job
EA20	Euro Area 20
EU	European Union
EU-LFS	European Union Labour Force Survey
Eurostat	Statistical Office of the European Union
ILO	International Labour Organization
ILOStat	International Labour Organization Employment Status
OECD	Organisation for Economic Co-operation and Development

Executive Summary

Average actual hours worked (AAHW) have declined across Nordic countries over the past decade, reducing aggregate labor input even as employment rates remained high. This matters for potential output, medium-term growth, and the assessment of labor market strength, since labor supply depends not only on the number of workers but also on the number of hours they supply.

This paper shows that the decline in AAHW in Denmark, Finland, Norway, and Sweden is driven mainly by reductions in hours worked among workers with low and medium levels of education. By contrast, highly educated workers tend to work longer hours and their increasing share in the workforce helps mitigate the aggregate decline. This offset is mechanical as well as economic: the decomposition weights each group by its share in employment, so even a small positive contribution from a large and growing high-skilled group can partly counteract larger within-group hours declines among lower-skilled workers. Decompositions including additional dimensions further show that this result remains visible when education is interacted with age, gender and employment status, with the negative contributions of lower- and medium-educated workers not confined to older workers, and that citizenship splits mainly change the magnitude rather than the direction of the core education pattern. Finland is a partial exception in the post-pandemic period, when hours worked among highly educated workers declined more noticeably, although their contribution remains positive over the full 2011–2022 period.

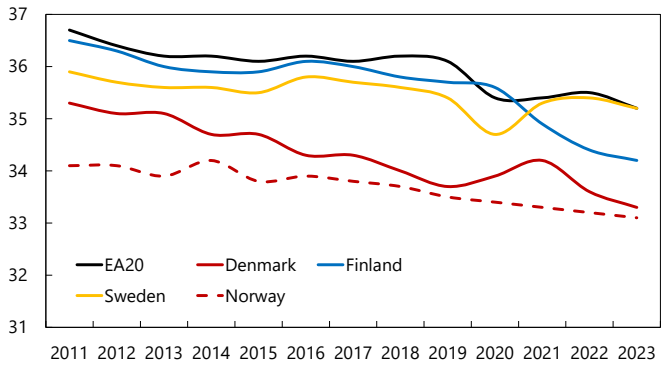
These findings suggest that declining hours worked in Nordic economies are closely linked to skill composition and to weaker hours performance among low- and mid-skilled workers, rather than to a generalized reduction in hours across all groups. They also suggest that age, gender, full-time/part-time status, and citizenship matter primarily as sources of heterogeneity around an education-centered result, not as alternative explanations that displace it. Because the analysis is descriptive, these findings identify areas for policy attention—including skills, job quality, and transitions into more productive employment—rather than establishing which interventions would increase hours worked.

Supplementary evidence on desired versus usual hours suggests that changing preferences alone may not fully explain the decline, but does not identify the relative roles of worker incentives, employer demand, and job-allocation frictions. Lower-educated workers more often appear to want to work more hours, while tertiary-educated workers are generally closer to their preferred hours and in some countries appear to want fewer. This pattern may reflect underemployment or other constraints on realized hours, stronger incentives to seek additional hours, or both; the descriptive evidence does not distinguish among these mechanisms.

Introduction

Average actual hours worked (AAHW) have declined across many advanced economies over recent decades. While rising productivity and improvements in living standards have long been associated with reductions in working time, the recent decline in hours worked has attracted renewed attention among policymakers and researchers. Because aggregate labor input depends not only on the number of workers but also on the number of hours supplied by each worker, changes in AAHW can have important implications for potential output, economic growth, and labor market performance.

Figure 1. Average Number of Actual Weekly Hours of Work in Main Job (Hours)



Sources: Eurostat; and IMF staff calculations.

Nordic countries provide a particularly interesting setting in which to study the evolution of hours worked. Denmark, Finland, Norway, and Sweden combine high labor force participation rates with relatively low levels of average hours worked compared with other advanced economies. At the same time, these economies have experienced important structural changes in their labor markets over the past decade, including rising educational attainment, changes in workforce composition, and evolving work arrangements. Understanding the drivers of declining hours worked in Nordic economies is therefore important for assessing medium-term labor supply dynamics and growth prospects.

A growing literature documents that the long-term decline in hours worked across advanced economies reflects both structural and institutional factors. These include demographic changes, technological progress, labor market regulations, and shifts in worker preferences. However, much less attention has been devoted to the role of changes in workforce composition—particularly changes in educational attainment—in shaping the evolution of hours worked. Recent European evidence also points to meaningful heterogeneity by gender, work arrangements, and migration-related characteristics, suggesting that workforce composition matters along several margins at once (Astinova et al., 2024). Over the past decades, most Nordic countries have experienced a substantial increase in the share of highly educated workers. Because highly educated workers tend to work longer hours on average, this compositional shift could potentially support aggregate hours worked even as other structural forces reduce working time.

This paper examines the evolution of AAHW in Nordic countries between 2011 and 2022, with particular emphasis on the role of education. Using microdata from the European Union Labour Force Survey (EU-LFS), the analysis first documents stylized facts on the evolution of hours worked across worker groups. It then applies a Shapley decomposition framework to quantify the contribution of changes in workforce composition and changes in hours worked within specific groups. To connect the education result to the broader labor-

supply literature, the paper also reports supplementary decompositions by gender, age, full-time/part-time status, and citizenship.

The results indicate that the decline in AAHW in Nordic countries is primarily driven by reductions in hours worked among workers with low and medium levels of education. Across all four countries, the largest negative contributions to the decline in hours worked originate from workers with primary and secondary education levels. By contrast, highly educated workers tend to work longer hours on average and their increasing share in the workforce exerts a mitigating effect on aggregate hours worked. Finland represents a partial exception during the pandemic and post-pandemic period, when hours worked among highly educated workers declined more noticeably. However, over the full sample period, tertiary-educated workers still contribute positively to aggregate hours worked in all Nordic countries. Additional results show that the education pattern remains robust once groups are further split by age, gender and employment status; in particular, the negative contributions from lower- and medium-educated workers span multiple age groups rather than being confined to older workers, while citizenship decompositions suggest that most of the aggregate variation is concentrated among citizens and that non-citizen contributions are usually smaller and sometimes offsetting.

These findings highlight the importance of skill composition in shaping labor input. While rising educational attainment tends to support aggregate hours worked, reductions in hours among workers with lower and medium levels of education can offset these gains and contribute to a decline in average hours worked. Understanding these dynamics is therefore important for identifying areas for further policy analysis, improving labor market outcomes, and sustaining long-term economic growth.

Additional evidence on desired versus usual hours indicates that a purely preference-driven interpretation is incomplete, but does not establish whether the gaps reflect employer-side constraints, stronger worker demand for hours, or both. Less-educated workers often report positive gaps between desired and usual hours, whereas tertiary-educated workers are generally closer to their preferred hours and in some cases report wanting fewer hours.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature on hours worked and labor supply in advanced economies. Section 3 describes the institutional context relevant for interpreting hours worked in Nordic labor markets. Section 4 presents stylized facts on the evolution of AAHW in Nordic countries and across worker groups. Section 5 describes the methodology and data used in the decomposition analysis. Section 6 presents the decomposition results and discusses their implications. Section 7 examines supplementary evidence on desired versus usual hours. Section 8 concludes with policy-relevant considerations.

Literature Review

Long-Run Decline in Hours and the Intensive Margin of Labor Supply

A large literature documents a secular decline in hours worked per worker across advanced economies, with the intensive margin accounting for much of the long-run change in aggregate labor input. Recent evidence underscores hours-per-worker as a key determinant of potential output alongside employment and participation. Using data for the United States and 18 European countries, Bick et al. (2022) show that

aggregate hours per person declined in most countries prior to COVID-19, mainly reflecting lower hours per worker even as employment rates rose. Their framework highlights how changes in fixed costs of working and work arrangements can generate diverging movements between extensive and intensive margins.

For Europe, Astinova et al. (2024) document a broad-based decline in average actual hours worked (AAHW) since 2003. Within-group reductions account for the bulk of the aggregate fall, and declines are widespread across industries and occupations. The contraction is most pronounced among youth and men, patterns that are remarkably consistent across countries. By comparing actual, usual, and desired hours, the authors argue that falling hours largely reflect shifting preferences rather than rising constraints: desired hours declined alongside actual hours for key groups. This interpretation aligns with cross-country evidence linking higher GDP per capita to lower average hours, consistent with income effects shaping labor-leisure choices over time. For the four Nordic countries specifically, prior evidence using EU-LFS data for 1996–2016 documents divergence rather than convergence in working-hour trends: hours were relatively stable in Finland and Sweden, while declining more markedly in Denmark and Norway, with younger age groups playing a notable role (Riekhoff, Krutova and Nätti, 2019). The role of educational composition in driving these divergent trends, however, has not been systematically examined for the Nordic region. Identifying that contribution is the central empirical objective of the present paper.

Preferences, "Willingness to Work," and Changing Labor Supply among Youth and Lower-Educated Workers

Post-pandemic research increasingly examines self-reported preferences, desired hours, and reservation wages to distinguish labor supply shifts from demand factors. Faberman et al. (2022), using the New York Fed's Survey of Consumer Expectations, document a pandemic-era decline in willingness to work along both extensive and intensive margins, partly driven by lower desired hours. This complements European evidence that actual and desired hours moved together for groups driving long-run declines.

Several mechanisms appear particularly relevant for younger and less-educated workers. Aguiar et al. (2021) show that young men experienced substantial declines in market work alongside increased leisure—especially video gaming—illustrating how improved leisure technologies can depress labor supply. Evidence on reservation wages and job search behavior suggests that expectations and wage-setting dynamics shifted after 2020. Research using longitudinal job search data shows that workers tend to set initial reservation wages above the predictions of the canonical job search model and revise them more slowly than theory implies, suggesting that labor supply expectations—including preferences over job quality and working hours—can be persistent even in the face of changing market conditions (Davis and Krolikowski, 2024). New York Fed data from the Survey of Consumer Expectations show that real reservation wages rose in the post-pandemic period even after accounting for inflation, with the distribution compressing across education and employment-status groups—suggesting that workers with lower educational attainment raised their wage floors relatively more than higher-educated workers (Kosar et al., 2024).

The tight post-2020 labor market further altered outside options and job-to-job mobility. Autor et al. (2023) document rapid wage growth, compression, and strong reallocation—especially among young non-college workers—consistent with an environment where flexibility, amenities, and search frictions interact with preferences to shape hours supplied. Broader survey evidence from OECD and Eurofound (2022) highlights

increased demand for flexible or reduced working time among younger cohorts, reinforcing the structural interpretation of lower AAHW.

Education, Hours Worked, and Productivity: Mechanisms and Empirical Links

Education influences AAHW through multiple channels. Higher human capital raises potential wages and job quality, potentially increasing hours, but income effects and greater access to flexible arrangements may reduce desired hours. Education also interacts with schooling decisions among youth, mechanically lowering hours when part-time work is combined with study. Astinova et al. (2024) show that a rising share of young part-time workers enrolled in education explains part of the observed decline in hours.

A related strand of the literature suggests that higher education and skill-intensive occupations can help sustain hours worked, even as average hours decline over time. In European microdata, highly educated workers tend to work longer hours on average, although part of this relationship reflects occupational and industry composition (Astinova et al., 2024). Evidence from the United States similarly shows that long work hours are disproportionately concentrated among highly educated and high-wage workers (Kuhn and Lozano, 2008), while the returns to long hours appear especially strong in skilled occupations (Cortés and Pan, 2019). Taken together, these findings suggest that rising educational attainment may partly cushion the decline in aggregate labor input, even if post-pandemic reductions in hours among some highly educated groups have weakened that offset.

Taken together, the literature identifies substantial heterogeneity in hours worked across demographic and education groups, but the contribution of educational composition to divergent Nordic trends remains less well established. The next section sets out the institutional context relevant for interpreting the paper's cross-country decompositions.

Institutional Context in Nordic Labor Markets

Nordic labour market institutions shape both the level of average actual hours worked and the pattern of adjustment across worker groups. Three institutional features are particularly relevant for interpreting the decomposition results in this paper: the design of parental leave systems, the regulation and incidence of part-time employment, and the role of collective bargaining in working-time setting.

Parental Leave Design

All four Nordic countries operate generous parental leave systems, but their institutional design differs substantially across countries and over the sample period, with direct implications for measured hours. Norway introduced the world's first non-transferable father's quota in 1993; by 2018 this quota stood at 15 weeks out of a total leave entitlement of 49 weeks at full pay (Brandth and Kvande, 2020; Arnalds, Duvander and Lammi-Taskula, 2025). Sweden's system provides 480 days total per child, with 13 weeks non-transferable for each parent since 2016 (Arnalds et al., 2025). Denmark stood as a laggard in this dimension: a brief father's quota introduced in 1998 was abolished in 2002, and Danish leave policy prior to 2022 relied primarily on collective agreements rather than statutory individual entitlements to encourage paternal take-up (Arnalds et al., 2025). Finland's pre-2022 system included a six-week paternity leave period alongside a shareable parental leave

component, and was substantially reformed in 2022 to introduce a 16-week non-transferable parental leave quota for each parent.

These design differences are relevant to the paper's analysis because the EU-LFS records hours worked in the reference week, creating a systematic wedge between actual and usual hours for workers on leave. In Nordic countries, where leave take-up rates are high and leave duration varies substantially by educational group, this wedge is correlated with both gender and education. In Finland, this pattern is especially pronounced due to the availability of the Child Home Care Allowance (*kotihoidon tuki*, HCA), a benefit that allows parents to care for children at home until age three rather than returning to paid employment. Research shows that take-up of HCA is strongly inversely related to educational attainment: highly educated parents use shorter HCA periods and return to paid employment sooner, while lower-educated mothers account for a disproportionate share of longer HCA spells (Saarikallio-Torp et al., 2024; Salmi and Närvi, 2017). As of 2016, women constituted 92 percent of HCA recipients (Miettinen and Saarikallio-Torp, 2021). Astinova et al. (2024) document more broadly that parental leave is a material contributor to the gender gap in actual hours in European microdata. As described in the data note in Section 5, the present analysis follows standard Eurostat methodology and excludes zero-hour reference weeks, so leave-induced absences enter the analysis through the participation margin rather than through measured AAHW directly.

Part-Time Employment and Involuntary Underemployment

Part-time employment patterns differ meaningfully across the four countries in ways that interact with the decomposition results. Norway has a relatively high incidence of female part-time employment compared with its Nordic peers, and part-time contracts are especially widespread in the female-dominated health and social services sector (OECD, 2023; Nätti and Nergaard, 2014). Norwegian legislative reforms in 2006 and 2013 aimed to reduce involuntary part-time by making it easier for employees to claim expanded hours, but evidence suggests that workplace-level staffing practices in the health-care sector have continued to sustain extensive part-time work, especially among lower-qualified care workers (Kavli, Nicolaisen and Trygstad, 2019). This institutional feature is directly relevant to the decomposition results: part of the large negative contribution of lower- and medium-skilled workers in Norway may reflect structurally short working arrangements in health and social services rather than a purely preference-driven reduction in hours. Denmark's flexicurity model—combining flexible hiring and dismissal rules with income support and active labour market policies—creates a different adjustment mechanism, in which labour market adjustment often occurs through turnover and activation (Madsen, 2006; Andersen and Svarer, 2007).

Collective Bargaining and Working-Time Norms

Working-time norms in Nordic countries are shaped primarily through collective bargaining rather than statutory regulation alone. Trade union density remains high by international standards across all four countries, though it varies: as of 2019, density stood at approximately 68 percent in Denmark, 68 percent in Sweden, 60 percent in Finland, and 50 percent in Norway, with Denmark, Sweden and Finland benefiting from the Ghent system of union-administered unemployment insurance that sustains membership (OECD, 2024; Kjellberg, 2025). Collective bargaining coverage is near-universal across all four countries (OECD, 2024). Within this institutional framework, Finland introduced a substantially revised Working Hours Act (Act 872/2019), which entered into force on 1 January 2020. The reform introduced a new category of flexible working arrangement (*joustotyöaika*) allowing employees in specialist and expert roles to independently determine when and where at least half of

their working hours are performed, extended daily flexitime limits, and made working hours banks universally available across all workplaces regardless of whether a collective agreement was in place (Finnish Government, 2019). This structural change—taking effect at the onset of the pandemic—gave salaried specialist and expert workers in Finland greater discretion over when and where they work, enabling hour adjustments in either direction without requiring a new collective agreement. To the extent that some knowledge workers used this flexibility to reduce their hours—whether in response to pandemic-related disruptions to work schedules, increased remote work, or changed preferences—this reform may be one permissive institutional factor in the background of the post-2019 deviation in Finnish hours among highly educated workers documented in this paper.

Taken together, these institutional features provide contextual background for interpreting the decomposition results. Because the Shapley decomposition is an accounting exercise that attributes aggregate changes to proximate components without establishing causal mechanisms, the institutional discussion here is intended as interpretive framing rather than as evidence of the channels through which institutions operate. Country-specific working-time institutions—parental leave design, part-time norms, and collective bargaining arrangements—interact with education and gender in ways that vary across Denmark, Finland, Norway and Sweden, and that help motivate the cross-country comparison and the supplementary decompositions by gender, employment status, and citizenship.

Migration Status and Hours Worked in Nordic Countries

The Nordic labour markets have experienced substantial inflows of non-EU/EEA migrants alongside significant EU intra-mobility, resulting in a more diverse citizenship composition of the workforce over the sample period. Migrants and non-citizens tend to be overrepresented in lower-skill service occupations and are more likely to work under non-standard or part-time contracts, which may amplify compositional pressures on aggregate AAHW (Astinova et al., 2024; European Commission, 2023). At the same time, high-skilled migration—particularly in Norway and Sweden—has partly offset these effects by adding tertiary-educated workers to the labour force. This heterogeneity motivates the paper's supplementary citizenship decomposition as a robustness check on whether the education-based result is concentrated among citizens or reflects a broader pattern.

Desired hours and potential constraints on realized hours. Comparisons of desired and usual hours provide suggestive evidence on preference-driven adjustments and possible constraints on realized hours, but do not separately identify these mechanisms. A persistent positive gap between desired and usual hours may reflect underemployment or other frictions that constrain realized hours, stronger worker incentives to supply additional hours, or both. Gaps close to zero—or negative—indicate that workers are closer to, or above, their preferred schedules. In the Nordic context, these patterns may reflect the interaction of worker preferences and income incentives with work-scheduling practices, part-time arrangements, occupational structure, and access to stable full-time positions, particularly for less-skilled workers.

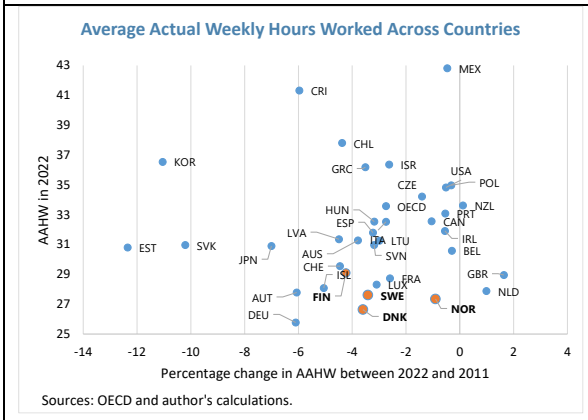
Overall, recent evidence suggests that declining AAHW in advanced economies reflects intensive-margin adjustments within demographic groups and, for key cohorts, evolving preferences rather than purely cyclical factors. In Nordic countries—characterized by high educational attainment, strong institutions, and generous family policies—education likely matters through both compositional shifts toward tertiary attainment and within-

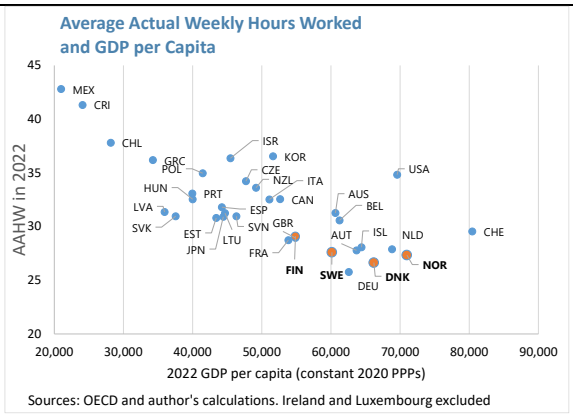
group hour changes, especially among lower-educated youth. This motivates the paper's empirical strategy of quantifying education's contribution to AAHW trends before and after 2019 and interpreting the results through the lens of preferences, schooling-work complementarities, and institutional settings.

Stylized Facts

Global Perspective

Figure 2. Average Actual Weekly Hours Worked





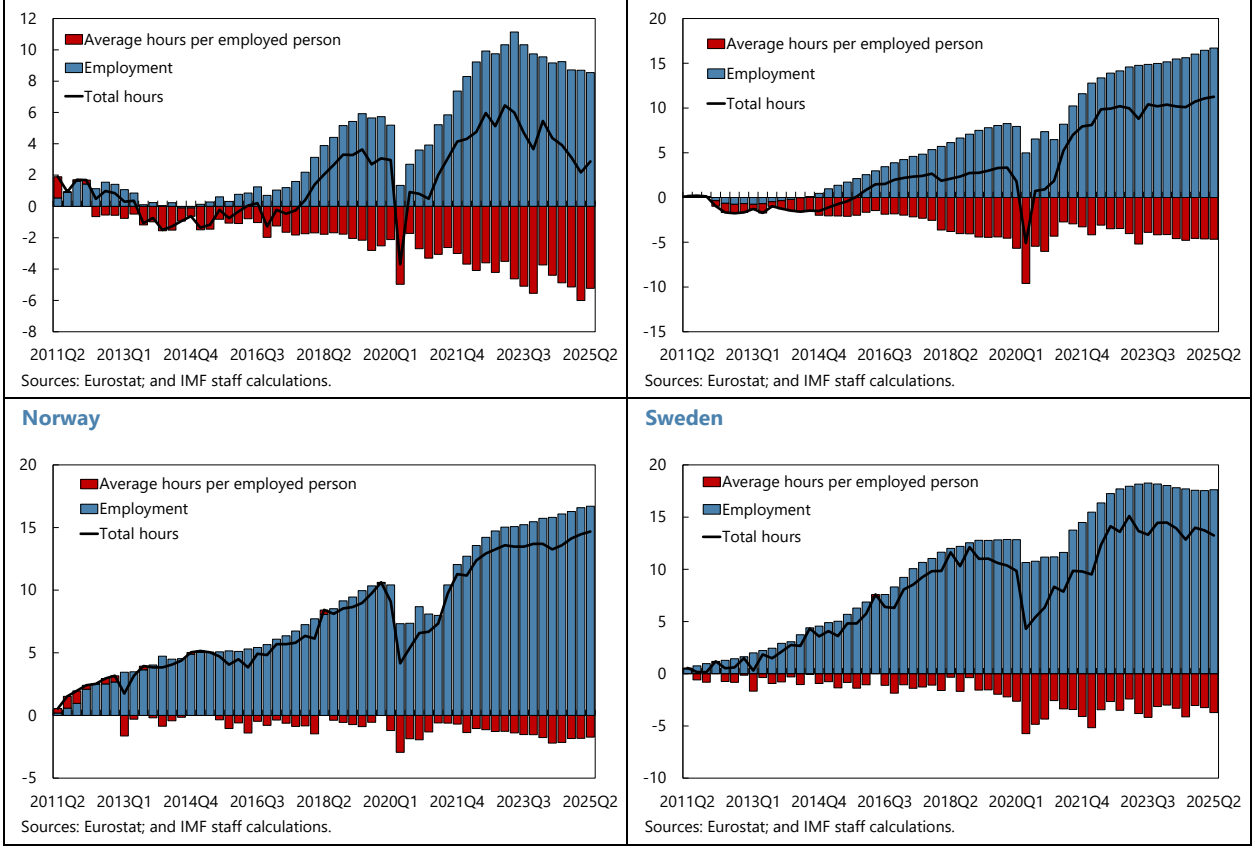
Average actual hours worked vary widely across countries and tend to decline as economies become richer. Figure 2 (left chart) illustrates the distribution of AAHW across a broad set of countries and the change in hours worked over the past decade. Nordic countries appear among the high-income economies with relatively low levels of AAHW. At the same time, the figure shows that many advanced economies have experienced declines in hours worked during this period, suggesting that the downward trend reflects broader structural developments rather than a phenomenon specific to Nordic labor markets.

The relationship between hours worked and income levels is further illustrated in Figure 2 (right chart), which plots AAHW against GDP per capita. Countries with higher income levels generally exhibit lower average hours worked, consistent with the long-standing observation that labor supply tends to decline along the intensive margin as productivity and living standards rise.

Importance for Total Labor Input

Changes in AAHW are important because they directly affect total labor input in the economy. While employment and labor force participation are often used as indicators of labor market performance, aggregate labor input depends both on the number of workers and on the number of hours each worker supplies. A decline in AAHW therefore reduces total hours worked even when employment levels remain stable or increase. Figure 3 illustrates this relationship for Nordic countries, showing that movements in AAHW contribute significantly to the evolution of total hours worked since 2011. In several countries, the growth of employment has partly offset the reduction in hours per worker, but declining AAHW nonetheless exerts downward pressure on total labor input. This highlights the importance of analyzing developments in hours worked when assessing labor supply dynamics and potential output.

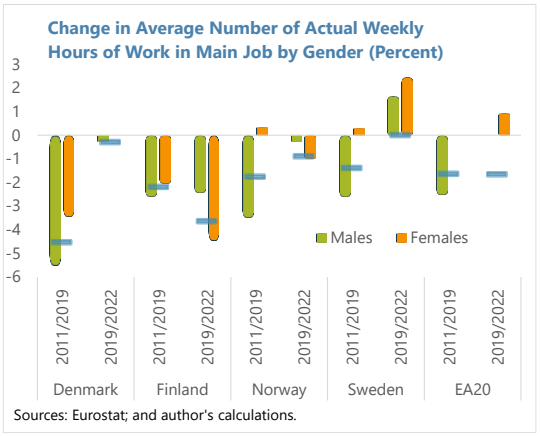
Figure 3. Total Hours Worked (Percent change since 2011Q1)



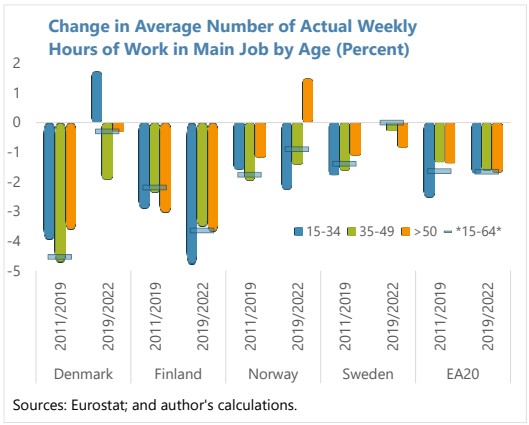
Heterogeneity across Worker Groups

The decline in AAHW also reflects heterogeneous developments across worker groups. Figure 4 presents changes in hours worked by gender, age group, and employment condition. While hours worked have declined across most categories, the magnitude of the decline varies across groups. In particular, the reduction in hours appears more pronounced among younger workers and among certain employment categories, suggesting that changes in labor market participation patterns and work arrangements may play an important role.

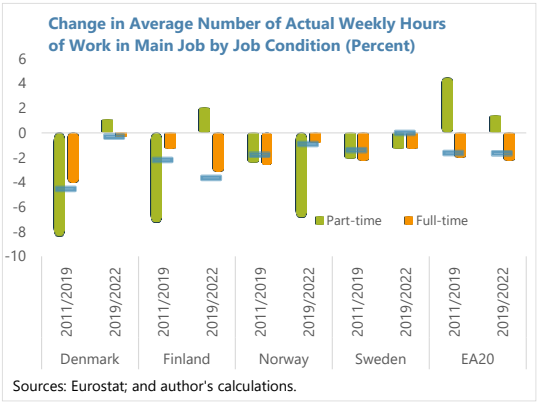
Figure 4. AAHW Changes by Worker Group



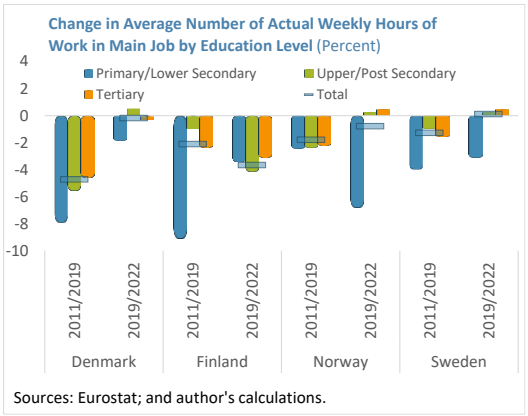
Sources: Eurostat; and author's calculations.



Sources: Eurostat; and author's calculations.



Sources: Eurostat; and author's calculations.



Sources: Eurostat; and author's calculations.

The Role of Education

Educational attainment plays a central role in shaping both the level and evolution of hours worked. Figure 5 shows average weekly hours worked by education group together with the share of workers in each group. Across all Nordic countries, workers with tertiary education consistently work longer hours on average than workers with lower educational attainment. Workers with primary or lower-secondary education typically report the lowest number of hours worked.

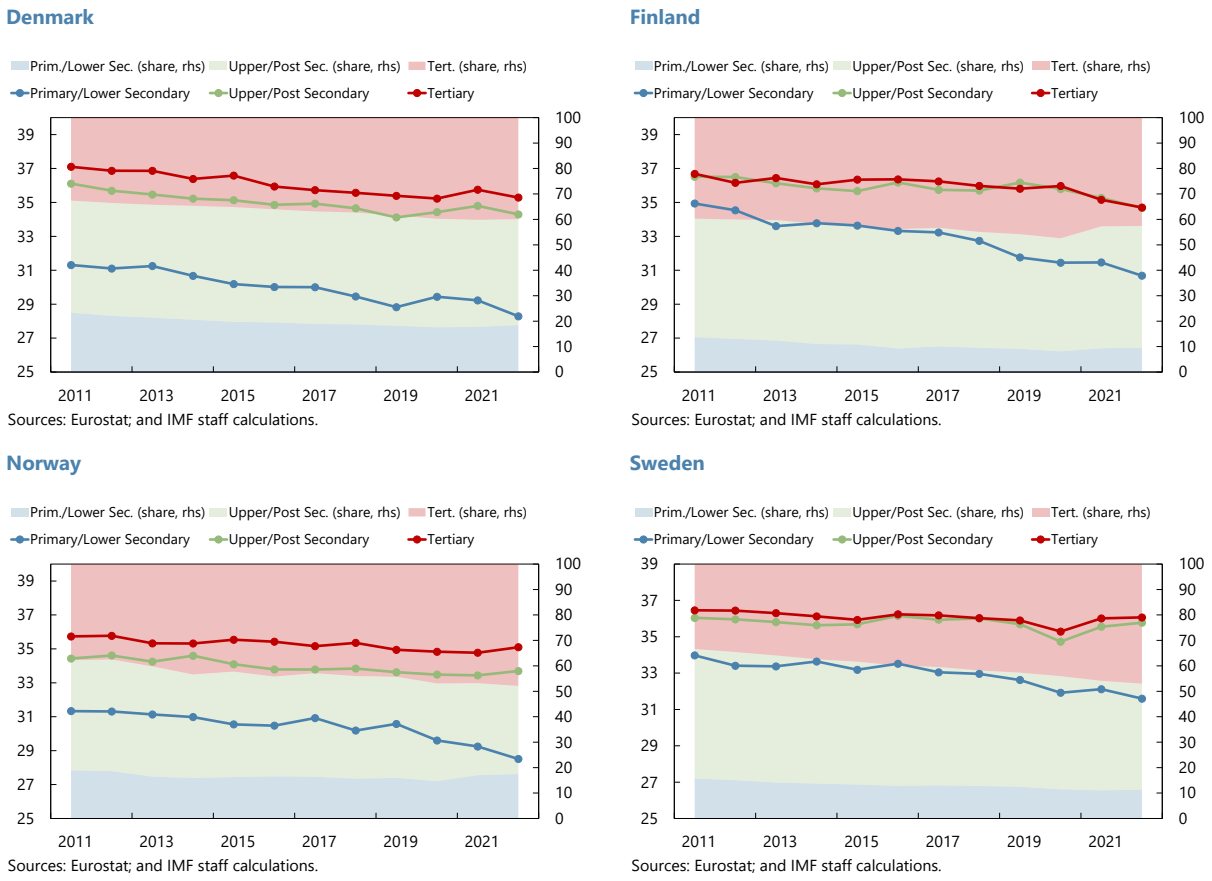
The figure also illustrates a steady increase in the share of tertiary-educated workers over time, accompanied by a decline in the share of workers with lower educational attainment. This shift toward higher education would mechanically support aggregate hours worked, given that highly educated workers tend to work longer hours on average.

However, the decline in AAHW observed in Nordic countries indicates that changes within education groups also play an important role. Descriptive evidence suggests that reductions in hours worked are more pronounced among workers with lower and medium levels of education, whereas hours worked among tertiary-educated workers remain comparatively stable. Finland shows a somewhat different pattern in the pandemic and post-pandemic period, when hours worked among highly educated workers declined temporarily.

Nevertheless, over the full sample period the increasing presence of highly educated workers continues to exert upward pressure on aggregate hours worked.

These patterns suggest that both compositional changes in the workforce and within-group dynamics may shape the evolution of AAHW. Distinguishing between these effects is therefore essential to understand the drivers of declining hours worked. The next section quantifies the relative contribution of these factors using a decomposition framework.

Figure 5. Actual Weekly Hours of Work in Main Job by Education Level
(Average in hours and share of workers in percent)



Methodology

Decomposition Framework

To quantify the contributors of changes in average actual hours worked, the paper uses a decomposition approach based on the Shapley value methodology (Shorrocks, 2013). This framework allows the total change in AAHW to be decomposed into contributions from specific worker groups, where each group's contribution reflects the combined effect of changes in its workforce share and changes in the hours worked by its members. The approach adapts a method originally developed for the analysis of poverty and inequality

(Barros et al., 2006; Azevedo, Sanfelice and Nguyen, 2012) to the context of average hours worked, exploiting the additivity property of a weighted mean to construct counterfactual distributions.

Let H denote average actual hours worked in the economy. This measure can be expressed as the weighted average of hours worked across worker groups:

$$\widehat{H} = \sum_i s_i h_i$$

where s_i represents the share of workers in group i, and h_i denotes the average hours worked by that group.

This formulation makes the weighting explicit: each group enters aggregate AAHW in proportion to its employment share. The decomposition diagrams therefore do not show unweighted changes in group-specific hours. They show each group's contribution to the aggregate change, combining the change in that group's average hours with the size and evolution of that group in the workforce. A small movement in hours for a large or rapidly growing group can therefore generate a sizable aggregate contribution, while a large hours change for a small group may have a limited effect on the aggregate.

The total change in AAHW between two periods can therefore be written as:

$$\Delta\widehat{H} = \sum_i \varphi_i$$

where φ_i is the Shapley-Shorrocks contribution of group i, which combines the composition effect (changes in that group's workforce share) and the within-group effect (changes in the hours its members work). These two components are analytically distinct—a group can contribute negatively because its share grew while its hours fell, because its share fell while it worked long hours, or through some combination—but the paper reports their sum φ_i rather than each component separately, since the object of interest is the net contribution of each group to the aggregate change. Separating these components could nevertheless provide useful additional insights in country-specific analysis, where differences in workforce composition and within-group hours dynamics can be linked more closely to national institutions and tailored policy questions.

When worker groups are defined along a single dimension—education level—the index i runs over the three education categories. When groups are defined by interacting two or more dimensions, the index i runs over all cells formed by that interaction. For example, interacting education (three levels) with gender (two categories) and employment status (full-time vs. part-time) produces twelve mutually exclusive cells, each with its own combined Shapley contribution φ_i. The cross-dimensional decompositions reported in Annex I use this cell-based approach to verify that the education gradient observed in the main results is not an artefact of a correlated dimension—for instance, that the contribution attributed to low-skilled workers is not simply a proxy for gender or part-time employment patterns.

Because the Nordic workforce is highly educated, the high-skilled group receives substantial weight in the aggregate decomposition. Its positive contribution therefore reflects both higher average hours and its rising employment share. Conversely, the negative contributions of low- and mid-skilled groups reflect not only weaker within-group hours performance but also their remaining weight in employment. This distinction is

important for interpreting the diagrams: the bars are contributions to aggregate AAHW, not simple within-group hour changes.

Path-Dependence and the Shapley Solution

A central challenge in sequential decomposition methods is path-dependence: the estimated contribution of each component depends on the order in which components are substituted from one period to the next. Different orderings of the same components can yield materially different results, making the decomposition sensitive to an arbitrary choice (Shorrocks, 2013). The Shapley value provides a principled solution to this problem. Rather than fixing a single ordering, it calculates the contribution of each component across all possible orderings and takes the average. These averages—known as Shapley-Shorrocks estimates—are unique, symmetric, and sum exactly to the total observed change, making the decomposition both internally consistent and path-independent (Shorrocks, 2013). The decomposition implemented constructs counterfactual distributions of AAHW by substituting observed values of each component one at a time across all possible orderings, then computes and averages the resulting counterfactual changes.

Worker Groups and Dimensions

The analysis focuses primarily on education because changes in educational attainment represent a key structural feature of Nordic labor markets and are most relevant for policy discussions. Gender, employment status, and citizenship status are included to capture additional heterogeneity in labor supply patterns; their contributions are reported in full in Annex I. In particular, the citizenship split helps assess whether the education-based result is concentrated among nationals or generalised across the broader workforce. Age is also interacted with education in the supplementary decompositions, which helps assess whether the education gradient reflects education-specific hours dynamics rather than only demographic composition within education groups.

Data and Sample Construction

The analysis uses microdata from the European Union Labour Force Survey (EU-LFS) for Denmark, Finland, Norway, and Sweden. The EU-LFS provides harmonised quarterly data on employment, hours worked, educational attainment, and demographic characteristics across EU and EEA member states. The sample covers the period 2011–2022 and is restricted to employed persons of working age.

Workers reporting zero hours in the reference week are excluded from the computation of average actual hours worked. This follows standard Eurostat methodology and ensures that the paper's microdata-based AAHW estimates remain consistent with the Eurostat-based series used for cross-country comparisons in the Stylized Facts section. As a result, the decomposition captures hours worked conditional on positive observed weekly hours rather than an unconditional average across all employed persons. An implication of this choice is that the analysis does not capture the contribution of zero-hour weeks to aggregate labor input. If lower- or medium-educated workers experience increasingly frequent or longer zero-hour spells over the sample period, this would affect total labor input through margins not reflected in the AAHW measure analyzed here. Disentangling the role of absence spells or leave intensity from within-employment hour dynamics would require complementary analysis of leave take-up and absence frequencies by education group, which is left for future research.

The supplementary desired-versus-usual hours series presented below is derived from the EU-LFS variable recording the number of hours employed persons would prefer to work per week. This variable is available in harmonised form across all four countries throughout the 2011–2022 sample period and is used here to compute annual averages of the gap between desired and usual hours by education group.

Sample Periods

The decomposition is performed for the full sample period (2011–2022). To distinguish between longer-term structural trends and pandemic-related adjustments, the analysis also examines two subperiods: the pre-pandemic period (2011–2019) and the pandemic and post-pandemic period (2019–2022). The pre-pandemic subperiod captures the structural drift in AAHW that was already underway before the COVID-19 shock. The post-pandemic subperiod isolates the additional adjustments that occurred from 2019 onward, including both the immediate pandemic-period disruption and the subsequent partial recovery through 2022.

Scope and Limitations

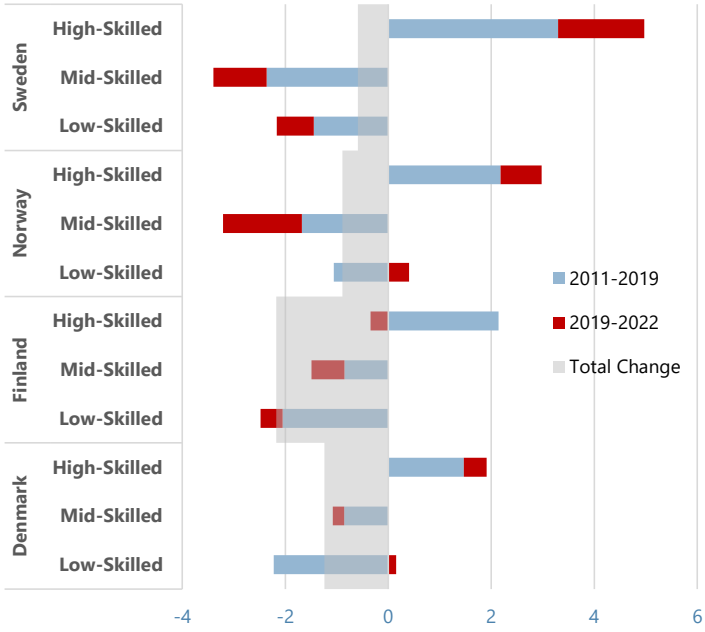
The decomposition identifies the proximate contributions of each worker group to the aggregate change in AAHW—as defined in the decomposition framework above—but does not identify the causal mechanisms behind those contributions. This is an inherent feature of the Shapley decomposition approach: by construction, the counterfactuals are not the result of an economic equilibrium but rather a statistical exercise in which one factor is modified at a time while others are held constant (Azevedo, Sanfelice and Nguyen, 2012). The results should therefore be interpreted as a consistent accounting of the proximate sources of the AAHW decline rather than as estimates of structural or causal parameters. Explaining why hours among low- and mid-skilled workers have declined, or identifying the policy interventions that would reverse this trend, would require complementary analysis using quasi-experimental variation or structural labour supply models.

The paper also does not directly estimate the relationship between wages, skill premia, and hours worked. Educational attainment is used as a proxy for skill levels, but the analysis does not observe task-specific skills, occupation-level productivity, or comparable hourly earnings over the full sample in a way that would allow a direct decomposition of wage incentives. Any interpretation of earnings-related mechanisms should therefore be viewed as suggestive and as a motivation for future work.

Decomposition Results

Overall Decomposition (2011–2022)

Figure 6. Decomposition of the Change in Average Actual Hours Worked



Source: Authors' calculations based on Eurostat EU Labour Force Survey (EU-LFS) microdata.
Note: Blue bars show cumulative changes in average actual hours worked over 2011–2019; red bars show cumulative changes over 2019–2022; gray bars show the total cumulative change over 2011–2022.
Note: Contributions are Shapley-Shorrocks contributions to the aggregate change in AAHW. They are weighted by each group's employment share and combine changes in group-specific average hours with changes in group composition. Education categories are used as proxies for skill levels.

The decomposition results confirm that the decline in average actual hours worked in Nordic countries is primarily driven by workers with low and medium levels of education. Figure 6 summarizes the education-related contributions to the change in AAHW between 2011 and 2022 for Denmark, Finland, Norway, and Sweden. Across all four countries, the largest negative contributions originate from workers with primary and lower-secondary education and, in most cases, from workers with upper-secondary education as well. By contrast, tertiary-educated workers contribute positively to aggregate AAHW in all countries over the full sample period.

This pattern is economically important because it shows that the decline in AAHW is not the result of a generalized reduction in hours across all education-based skill groups. Instead, the downward pressure on aggregate hours comes mainly from weaker hours performance among low- and mid-skilled workers, while tertiary-educated workers help sustain total labor input. The mitigating role of high-skilled workers is amplified

by the decomposition weights: because tertiary-educated workers account for a large and rising share of employment in Nordic countries, even modest resilience in their average hours can generate a meaningful positive aggregate contribution. Conversely, the negative contributions of low- and mid-skilled workers indicate that their within-group declines in hours have been large enough to dominate any offset from the changing workforce composition.

The cross-country comparison reveals meaningful differences in the relative weight of education groups. In Norway and Sweden, Figure 6 shows that the negative contribution of upper-secondary workers is especially pronounced—it is the single largest negative bar in both countries. In Denmark and Finland, by contrast, the downward pressure from workers with primary or lower-secondary education is comparatively stronger, and accounts for the dominant share of the aggregate decline. The Annex I decomposition, which further splits worker groups by gender and employment status, gives more precise magnitudes: in Norway, male mid-skilled full-time workers account for approximately −1.2 hours of the total change, compared with −0.9 hours for male low-skilled full-time workers. In Sweden, the corresponding figures are approximately −1.5 and −0.9 hours respectively. In Denmark and Finland, male low-skilled full-time workers are the single largest negative contributor, at approximately −0.8 and −1.2 hours respectively. Despite these cross-country differences in magnitude, the qualitative pattern is consistent: lower educational attainment is associated with negative contributions to AAHW, while highly educated workers exert a positive offsetting effect across all four countries.

Additional Decomposition Dimensions

The broader decomposition, reported in Annex I, shows that the education-based result remains robust when worker groups are further split by age, gender, employment status, and citizenship. The education-by-age results suggest that the education gradient is not confined to older workers: negative contributions from lower- and medium-educated workers are distributed across multiple age groups, while highly educated workers generally provide offsetting positive contributions, notwithstanding some post-pandemic exceptions. The employment-status results similarly reinforce the main finding. The largest negative contributions across all four countries tend to originate from low- and mid-skilled workers in full-time employment, while the contributions associated with part-time categories are generally smaller in magnitude. This is a notable finding: it suggests that the decline in AAHW is not primarily a part-time employment story. Rather, it reflects weaker hours dynamics among less-educated workers, including within full-time jobs.

The decomposition also indicates that gender and citizenship differences are relevant but secondary to the education dimension. In Norway and Sweden, male mid-skilled full-time workers account for the single largest negative contributions to the aggregate decline. In Denmark and Finland, male low-skilled full-time workers play the dominant negative role, though female mid-skilled full-time workers also make material negative contributions—approximately −0.5 hours in Denmark and −1.0 hours in Finland. High-skilled full-time workers of both genders contribute positively in all four countries, with especially large positive contributions in Sweden (male high-skilled full-time: approximately +1.5 hours; female high-skilled full-time: approximately +2.0 hours) and Finland (female high-skilled full-time: approximately +1.5 hours). The citizenship decompositions point in the same direction: most of the large negative and positive bars are concentrated among citizens, while non-citizen contributions are generally smaller and often positive before the pandemic. Post-pandemic exceptions are limited but informative—in Norway, non-citizen mid-skilled workers also make a visible negative contribution after 2019, while in Denmark and Finland non-citizen contributions are mostly offsetting. These differences across gender and citizenship do not alter the main conclusion: educational attainment remains the

most informative and policy-relevant dimension for understanding the decline in AAHW in Nordic economies. The citizenship results are suggestive and warrant further investigation, but caution is appropriate before drawing policy conclusions. The decomposition attributes contributions to citizenship groups as a whole; disentangling whether the non-citizen patterns reflect changes in the share of non-citizens in the workforce, changes in the hours non-citizens typically work, or changes in their skill and sector composition would require deeper analysis beyond the scope of this paper.

Pre-Pandemic Dynamics

The pre-pandemic period accounts for most of the structural decline in AAHW observed in Nordic countries. The decomposition for 2011–2019, shown in Figure 6, indicates that negative contributions from low- and mid-skilled workers were already well established before the pandemic. In all four countries, the blue bars representing the 2011–2019 subperiod account for the majority of the total negative contributions over the full sample. At the same time, the increasing share and relatively stable hours of tertiary-educated workers helped cushion the aggregate decline during this period. This indicates that the main mechanism emphasized in the paper—a negative contribution from lower-skilled workers partly offset by a positive contribution from highly educated workers—was already operating before 2020 and is not a pandemic-specific phenomenon.

Pandemic and Post-Pandemic Developments

The pandemic and post-pandemic period introduces some additional cross-country heterogeneity, but does not overturn the broader pattern. In Finland, Figure 6 shows that the 2019–2022 subperiod is marked by a visible negative contribution from tertiary-educated workers—the red bar for high-skilled workers extends to the left—making Finland the main within-region exception to the pattern of tertiary-educated resilience. This is a departure from the pre-pandemic period, in which Finnish highly educated workers contributed positively to AAHW. Even so, over the full 2011–2022 period, the cumulative contribution of tertiary-educated workers in Finland remains positive, as the stronger pre-pandemic performance of this group more than offsets the post-2019 weakening. The Finnish case therefore represents a post-pandemic deviation within a broader regional pattern rather than a structural break.

In the other three Nordic countries, the post-pandemic decomposition broadly reinforces the same message observed in the full sample. Figure 6 shows that the 2019–2022 red bars for lower- and medium-skilled workers remain negative in Denmark, Norway, and Sweden, while highly educated workers continue to contribute positively. The relative intensity of these contributions shifts somewhat—in some countries the post-pandemic period sees a moderation of the pre-pandemic negative contributions from low-skilled workers—but the directional pattern is unchanged. This suggests that the pandemic may have altered the pace or relative intensity of these developments without fundamentally changing the underlying structure of the decline in hours worked across Nordic labor markets.

Supplementary Evidence: Desired Versus Usual Hours

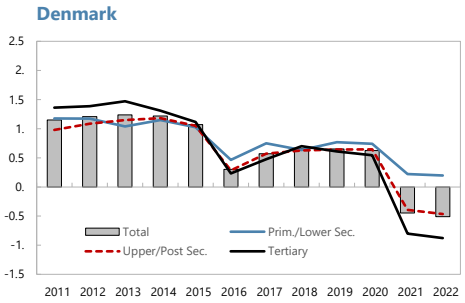
The decomposition results identify which groups account for the decline in AAHW and show that the pattern predates the pandemic. What they cannot resolve is whether the within-group hour reductions among less-

educated workers reflect changing preferences for shorter workweeks, stronger income-related demand for hours, binding constraints that prevent workers from obtaining their preferred hours, or a combination of these mechanisms. Lower desired hours would be consistent with a voluntary labor-supply adjustment, while positive desired–usual gaps may reflect constraints on realized hours, stronger income-related demand for hours, or both. These interpretations have different macroeconomic and policy implications. Preference-driven reductions in desired hours would bear more directly on effective labor supply and potential output, whereas persistent positive desired–usual hours gaps may signal labor-market slack or stronger worker incentives to supply hours; if the gaps reflect durable matching or job-allocation frictions, they could also weigh on potential output over time. The gap between desired and usual hours provides supplementary evidence on these possibilities but does not identify their relative importance.

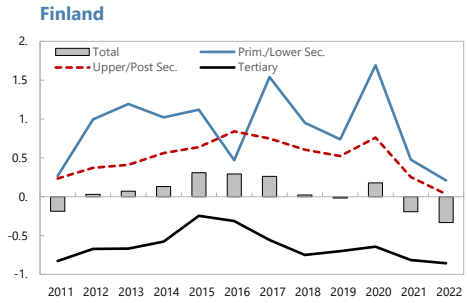
In Finland, Norway, and Sweden, workers with primary or lower-secondary education display the largest positive gaps through most of the sample, indicating that they would like to work more hours than they usually do. Upper- and post-secondary workers also tend to show positive gaps, though generally smaller. By contrast, tertiary-educated workers are much closer to zero and, in Finland—and later in Denmark and Sweden—often display negative gaps, suggesting that highly educated workers are closer to, or at times above, their preferred hours.

These patterns do not by themselves identify the source of the gap. Two mechanisms are consistent with a persistent positive desired–usual gap among less-educated workers: underemployment or constraints on realized hours—where workers want more hours but cannot secure them due to employer scheduling practices, limited job availability, occupational sorting into short-contract roles, or other labor market frictions affecting how hours are allocated—and income-driven hour demand, where lower wages generate stronger incentives to supply additional hours to meet income targets. Both mechanisms would produce the same qualitative pattern and cannot be distinguished from the desired-hours gap alone. A constraint-based interpretation is one possible explanation consistent with the decomposition result showing that the hour shortfall is concentrated within full-time employment as well as in part-time categories, and with the institutional evidence in Section 3 on the prevalence of involuntary short-contract work particularly in Norway's health-care sector. This interpretation complements rather than contradicts the broader European evidence reviewed in Section 2. Astinova et al. (2024) find that desired hours fell alongside actual hours for key groups across Europe, pointing to preferences as the dominant force overall. What the Nordic education gradient in desired–usual hours suggests is that, even within that preference-driven environment, less-educated workers may face institutional barriers, stronger incentives to seek more hours, or a combination of the two. If further analysis confirms that these gaps reflect constraints on realized hours, measures addressing relevant institutional and labor-market frictions could be warranted. If instead they mainly reflect stronger income-related demand for hours or other supply-side factors, the appropriate policy response would differ.

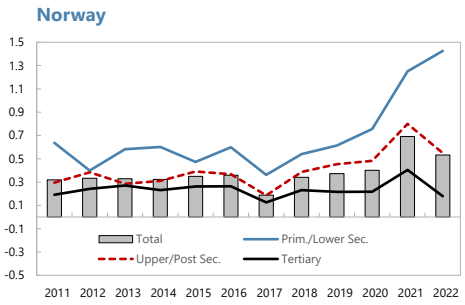
Figure 7. Gap between Desired and Usual Hours Worked by Education Level (Weekly hours)



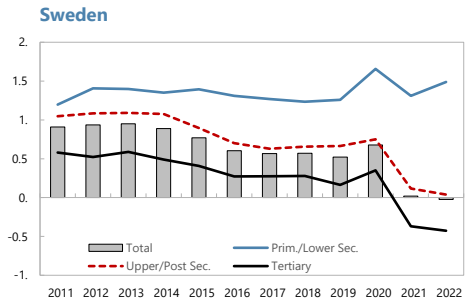
Source: Eurostat Labour Force Surveys and IMF staff calculations.



Source: Eurostat Labour Force Surveys and IMF staff calculations.



Source: Eurostat Labour Force Surveys and IMF staff calculations.



Source: Eurostat Labour Force Surveys and IMF staff calculations.

Concluding Remarks

Average actual hours worked have declined across Nordic countries over the past decade, reducing aggregate labor input even in economies with relatively high employment and labor force participation. Because hours worked are a key component of effective labor supply, this decline has important implications for potential output, medium-term growth, and the assessment of labor market strength.

The findings are particularly relevant in the context of population aging. As the share of older workers rises, aggregate labor supply will increasingly depend on whether longer working lives are accompanied by sustained hours, employability, and productivity. Higher skill levels may help buffer aging-related pressures by supporting employability, occupational mobility, and attachment to higher-quality jobs, but the decomposition does not establish a causal effect of skills on hours worked. Any such role should therefore be viewed as a possible partial buffer rather than a demonstrated offset to health, caregiving, or retirement-preference channels.

The decomposition results confirm that the decline is concentrated among workers with low and medium levels of education. Across all four countries, the negative contribution to AAHW is concentrated among workers with low and medium levels of education, while tertiary-educated workers contribute positively through both longer workweeks and a rising workforce share. Within this broad pattern, Denmark and Finland show larger negative contributions from primary/lower-secondary workers, whereas Norway and Sweden show larger negative contributions from upper/post-secondary workers. This pattern was already operating before 2020 and remains visible when the workforce is further split by age, gender, employment status, and citizenship; the age interaction indicates that the negative contributions of lower- and medium-educated workers are not confined to

older workers. Finland represents a partial exception in the pandemic and post-pandemic period, when highly educated workers also made a negative contribution, though their cumulative contribution over 2011–2022 remains positive. Because the decomposition is descriptive, these findings identify policy-relevant questions rather than the interventions that would reverse the decline in hours.

First, the positive contribution of tertiary-educated workers makes skills and educational upgrading relevant areas for policy consideration, but does not demonstrate that such policies would raise hours worked. Skills policies may nevertheless support productivity, adaptability, and employability on broader grounds. Looking ahead, structural changes in labour markets driven by technology and the green transition are increasing the premium on higher-order skills and lifelong learning, reinforcing the case for skill-upgrading policies (OECD, 2025).

Second, the results motivate closer examination of the constraints facing workers with lower and medium levels of education. Since these groups account for most of the decline in AAHW, measures that improve job quality, access to stable hours, and transitions into occupations with stronger labor demand may be relevant if positive desired–usual hours gaps reflect underemployment or job-allocation frictions. If lower hours instead reflect changing preferences, policies aimed primarily at increasing hours may be less effective or appropriate. Distinguishing between these mechanisms requires additional causal analysis.

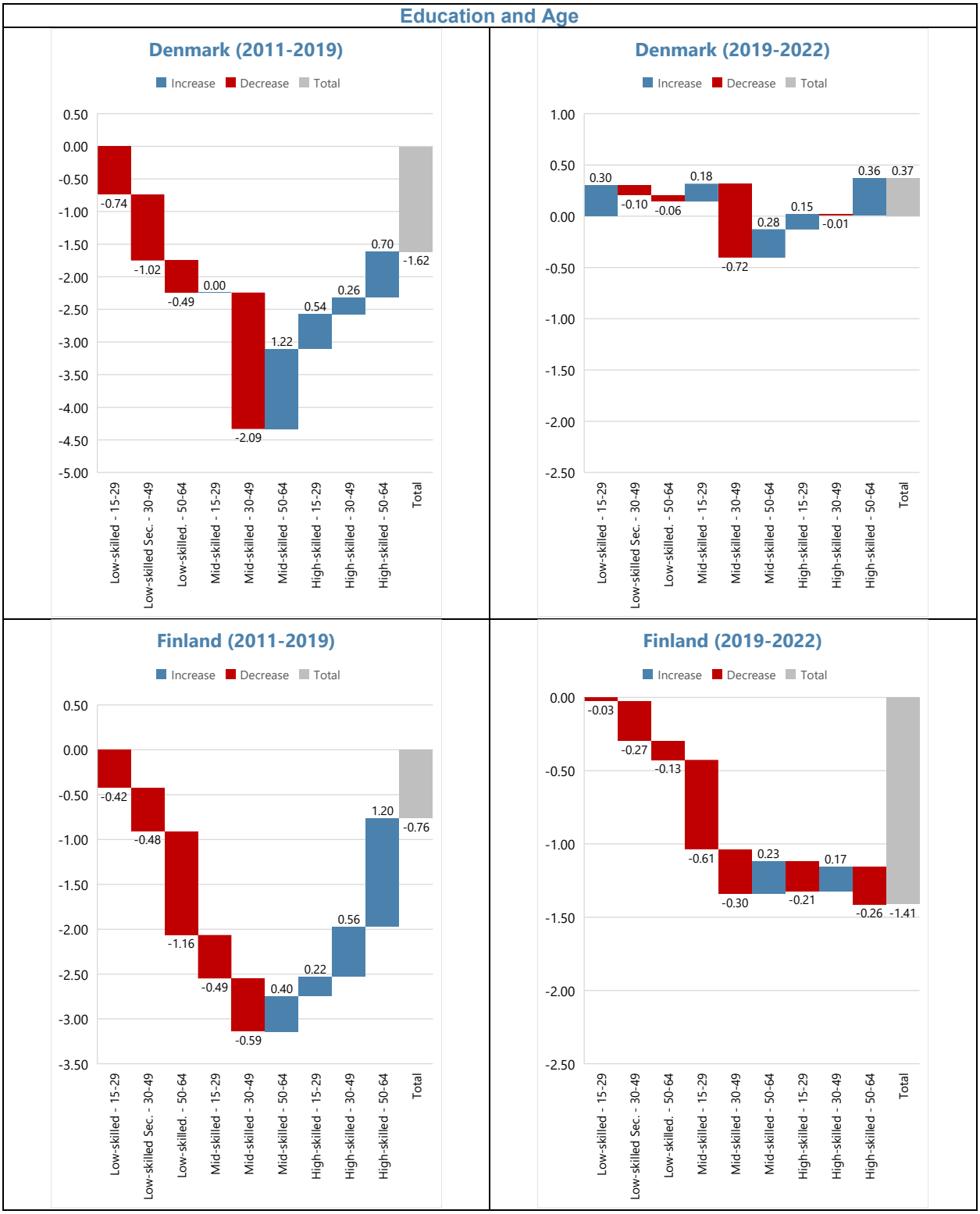
Third, additional decompositions confirm that the dominant descriptive pattern remains the weaker hours performance of less-educated workers—a pattern that holds within full-time employment and across genders. Gender and employment status are secondary dimensions that help characterize cross-country heterogeneity without displacing the education result. The citizenship decompositions point in a similar direction: most of the aggregate negative contributions are concentrated among citizens, and non-citizen contributions are generally smaller and often positive before the pandemic. This suggests that migration composition is not the primary driver of the AAHW decline, but the positive non-citizen contributions should not be interpreted as a straightforward policy lever either. The decomposition cannot separate whether this pattern reflects the hours non-citizens actually work versus their growing share in the workforce, or whether it varies by productive sector. Translating these results into migration policy conclusions would therefore require more targeted analysis.

The evidence on desired versus usual hours suggests that changing preferences alone may not fully explain the decline. Across much of the sample, lower-educated workers report larger positive gaps between desired and usual hours than tertiary-educated workers. These gaps may reflect constraints on realized hours, stronger income-related demand for hours, or both. If further analysis confirms that they reflect constraints on realized hours, measures addressing relevant institutional and labour-market frictions could be warranted. If instead they mainly reflect stronger worker incentives to seek additional hours or other supply-side factors, the appropriate policy response would differ.

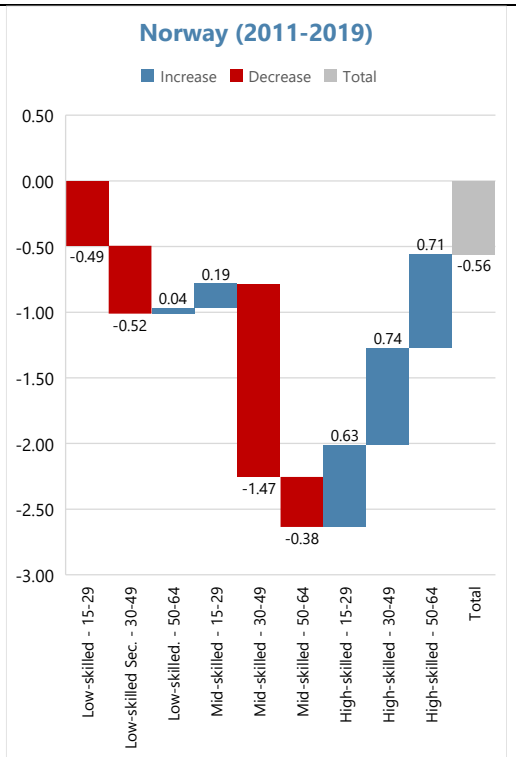
More broadly, the findings underscore the importance of monitoring labor input along the intensive margin. In countries where employment rates are already high, further gains in aggregate labor input may depend less on increasing the number of workers and more on understanding why some groups are working fewer hours over time. For Nordic economies, this points to the value of further analysis of the interaction between skills, workforce composition, and hours worked.

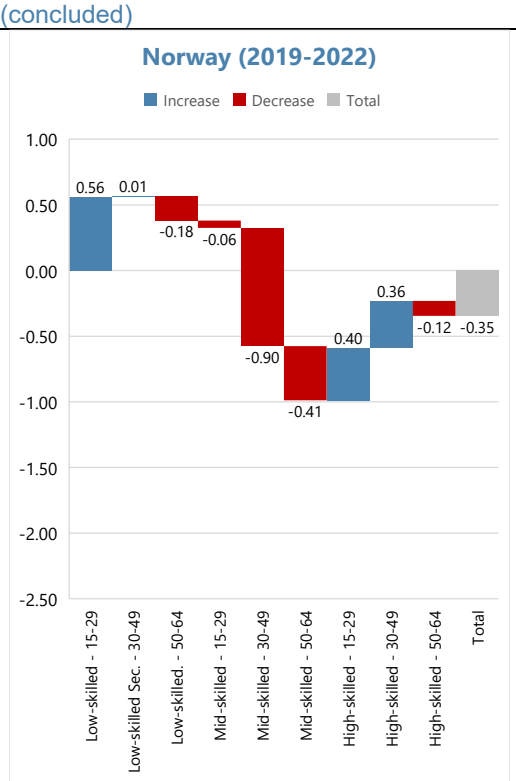
In sum, the decline in AAHW in Nordic countries reflects a persistent labor market pattern concentrated among low- and mid-skilled workers, although the descriptive evidence does not identify the relative roles of worker preferences and incentives, employer demand, and job-allocation frictions. Highly educated workers have partly cushioned this decline through both longer average workweeks and rising workforce shares. The supplementary decompositions by age, gender, employment status, and citizenship sharpen this interpretation but do not overturn it: they show that the education result remains the central organizing pattern, is not confined to older workers, with most citizenship-related variation concentrated among citizens and non-citizen contributions generally smaller. Understanding these offsetting forces is essential for interpreting labor input trends and for designing policies that support long-term growth.

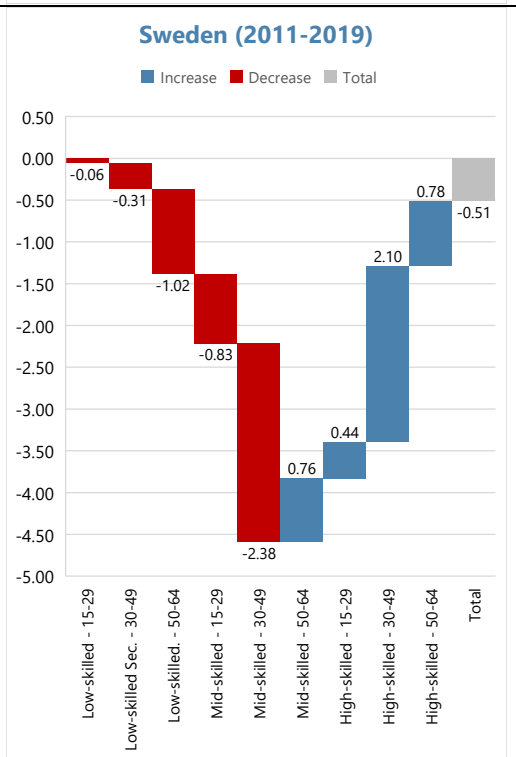
Annex I. Additional Shapley Decompositions

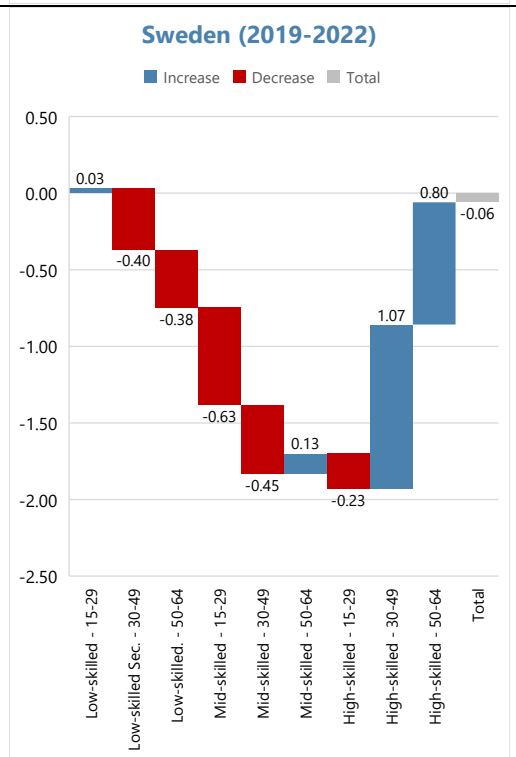


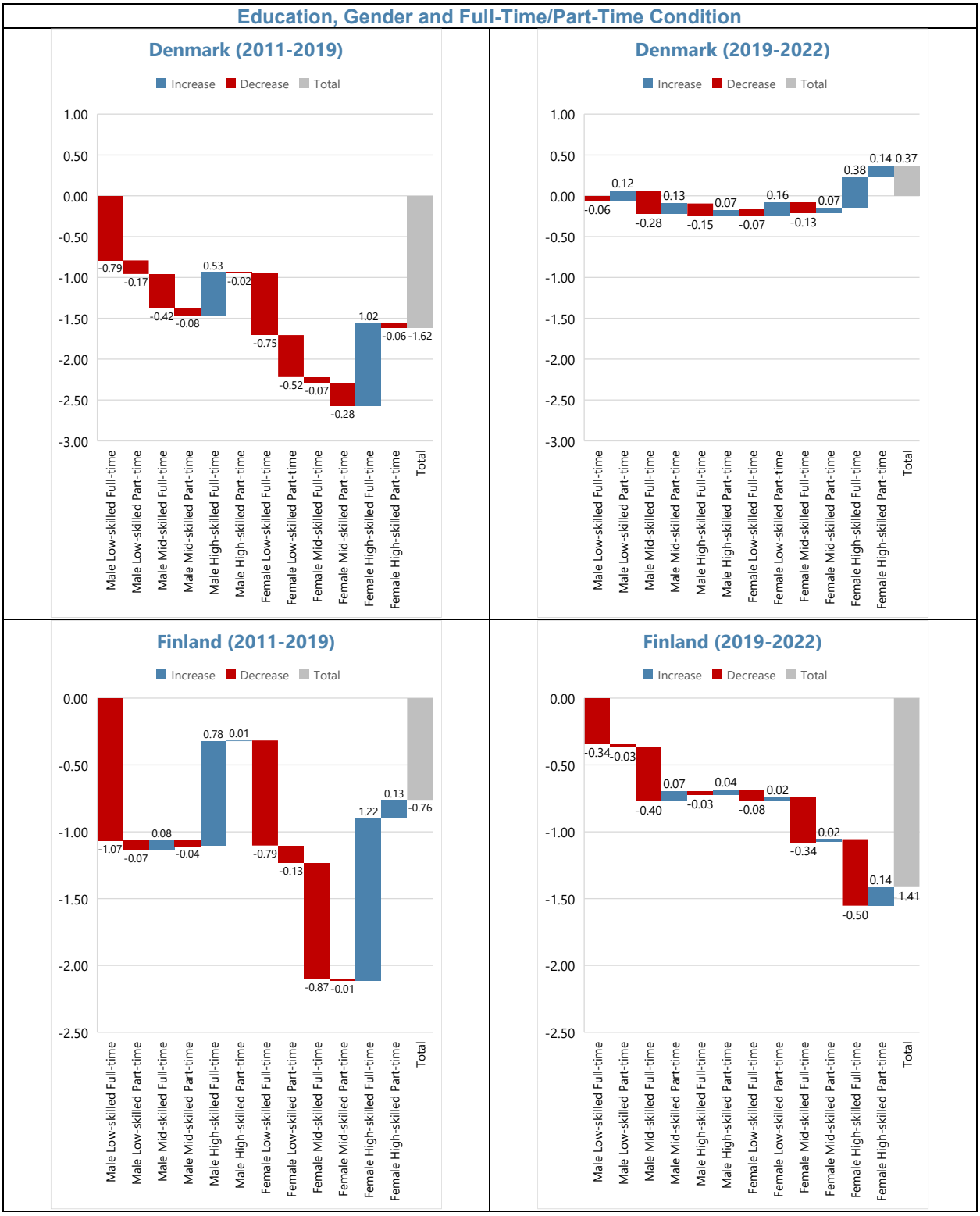
Education and Age (concluded)



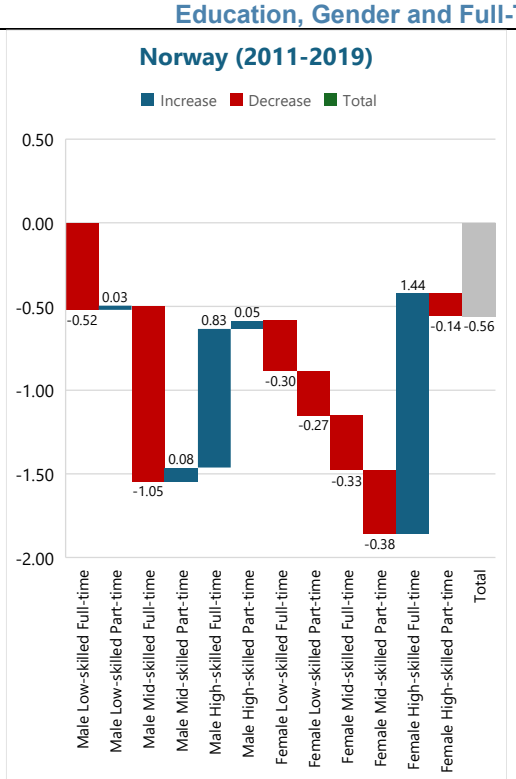


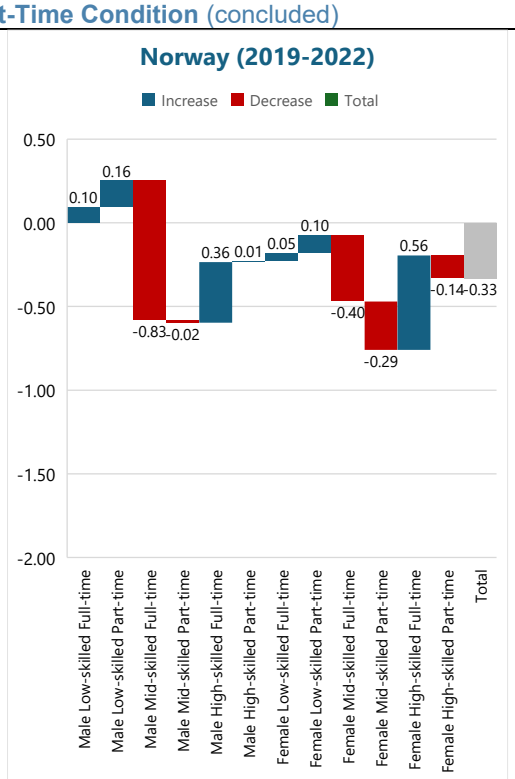


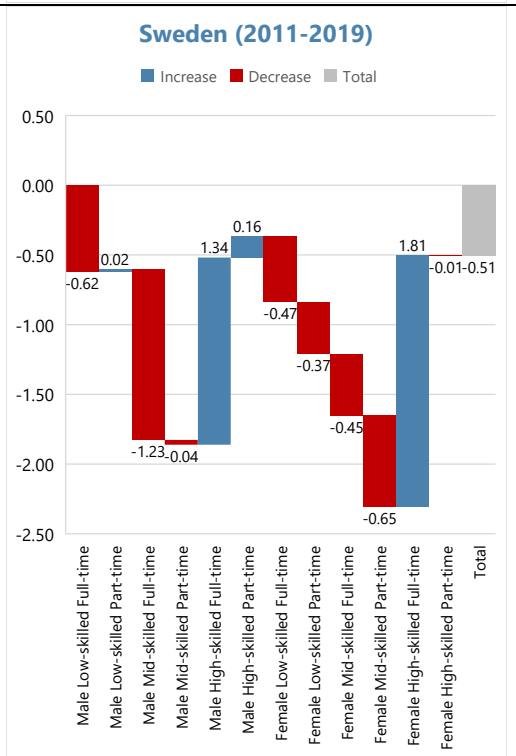


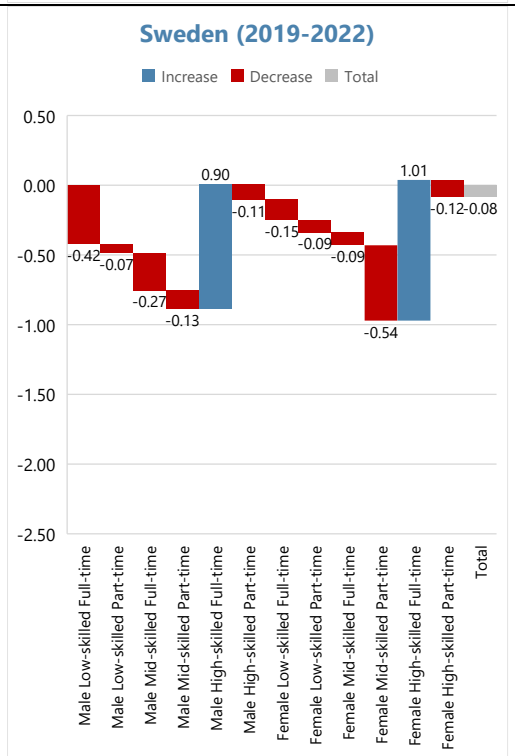

Education, Gender and Full-Time/Part-Time Condition
Denmark (2011-2019)
Increase
Decrease
Total
1.00
0.50
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
-3.00
-0.79
-0.17
-0.42
-0.08
0.53
-0.02
-0.75
-0.52
-0.07
-0.28
1.02
-0.06
-1.62
Male Low-skilled Full-time
Male Low-skilled Part-time
Male Mid-skilled Full-time
Male Mid-skilled Part-time
Male High-skilled Full-time
Male High-skilled Part-time
Female Low-skilled Full-time
Female Low-skilled Part-time
Female Mid-skilled Full-time
Female Mid-skilled Part-time
Female High-skilled Full-time
Female High-skilled Part-time
Total
Denmark (2019-2022)
Increase
Decrease
Total
1.00
0.50
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
-3.00
-0.06
0.12
-0.28
0.13
-0.15
0.07
-0.07
0.16
-0.13
0.07
0.38
0.14
0.37
Male Low-skilled Full-time
Male Low-skilled Part-time
Male Mid-skilled Full-time
Male Mid-skilled Part-time
Male High-skilled Full-time
Male High-skilled Part-time
Female Low-skilled Full-time
Female Low-skilled Part-time
Female Mid-skilled Full-time
Female Mid-skilled Part-time
Female High-skilled Full-time
Female High-skilled Part-time
Total
Finland (2011-2019)
Increase
Decrease
Total
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
-1.07
-0.07
0.08
-0.04
0.78
0.01
-0.79
-0.13
-0.87
-0.01
1.22
0.13
-0.76
Male Low-skilled Full-time
Male Low-skilled Part-time
Male Mid-skilled Full-time
Male Mid-skilled Part-time
Male High-skilled Full-time
Male High-skilled Part-time
Female Low-skilled Full-time
Female Low-skilled Part-time
Female Mid-skilled Full-time
Female Mid-skilled Part-time
Female High-skilled Full-time
Female High-skilled Part-time
Total
Finland (2019-2022)
Increase
Decrease
Total
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
-0.34
-0.03
-0.40
0.07
-0.03
0.04
-0.08
0.02
-0.34
0.02
-0.50
0.14
-1.41
Male Low-skilled Full-time
Male Low-skilled Part-time
Male Mid-skilled Full-time
Male Mid-skilled Part-time
Male High-skilled Full-time
Male High-skilled Part-time
Female Low-skilled Full-time
Female Low-skilled Part-time
Female Mid-skilled Full-time
Female Mid-skilled Part-time
Female High-skilled Full-time
Female High-skilled Part-time
Total

Education, Gender and Full-Time/Part-Time Condition (concluded)



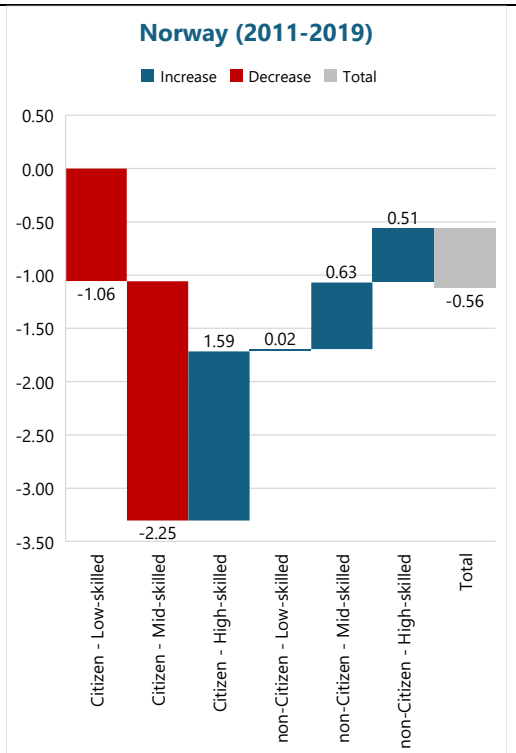


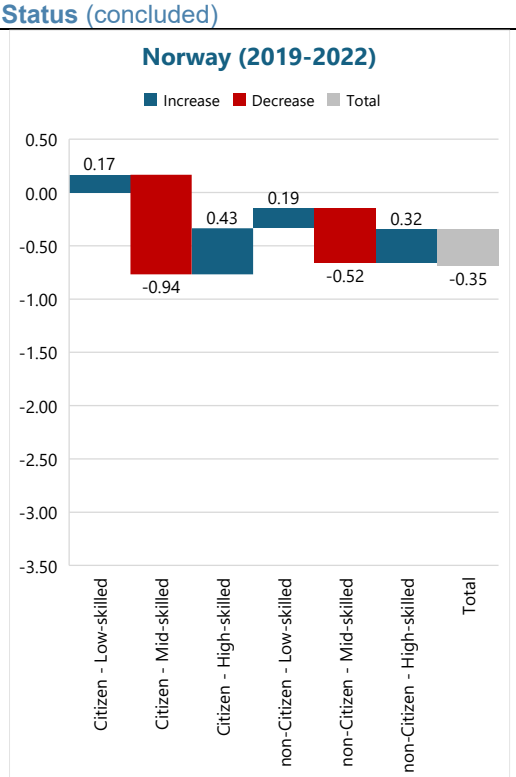


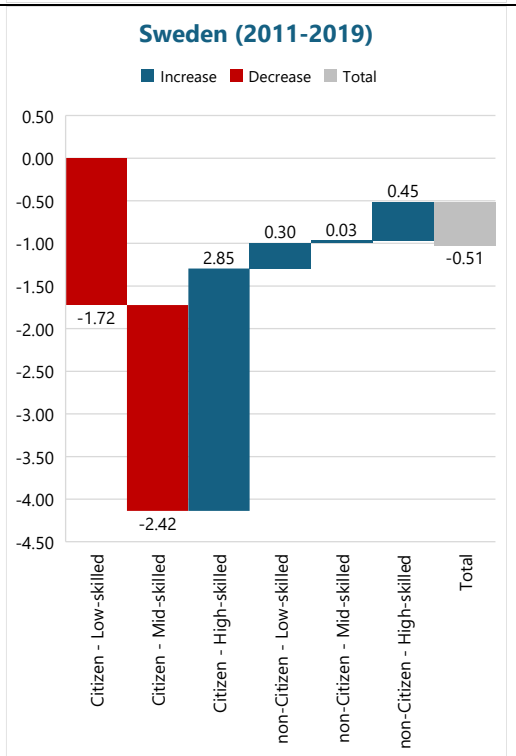


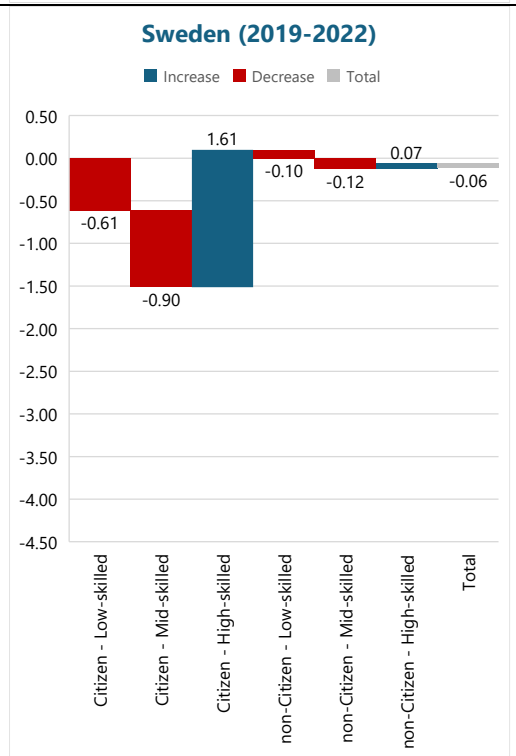

Education and Citizenship Status
Denmark (2011-2019)
Increase
Decrease
Total
1.00
0.50
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
-3.00
-3.50
-2.19
-0.83
1.09
-0.05
-0.02
0.39
-1.62
Citizen - Low-skilled
Citizen - Mid-skilled
Citizen - High-skilled
non-Citizen - Low-skilled
non-Citizen - Mid-skilled
non-Citizen - High-skilled
Total
Denmark (2019-2022)
Increase
Decrease
Total
1.00
0.50
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
-3.00
-3.50
-0.11
-0.50
0.31
0.24
0.26
0.17
0.37
Citizen - Low-skilled
Citizen - Mid-skilled
Citizen - High-skilled
non-Citizen - Low-skilled
non-Citizen - Mid-skilled
non-Citizen - High-skilled
Total
Finland (2011-2019)
Increase
Decrease
Total
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
-3.00
-3.50
-2.01
-1.04
2.01
0.00
0.12
0.15
-0.76
Citizen - Low-skilled
Citizen - Mid-skilled
Citizen - High-skilled
non-Citizen - Low-skilled
non-Citizen - Mid-skilled
non-Citizen - High-skilled
Total
Finland (2019-2022)
Increase
Decrease
Total
0.00
-0.50
-1.00
-1.50
-2.00
-2.50
-3.00
-3.50
-0.36
-0.91
-0.44
-0.12
0.21
0.21
-1.41
Citizen - Low-skilled
Citizen - Mid-skilled
Citizen - High-skilled
non-Citizen - Low-skilled
non-Citizen - Mid-skilled
non-Citizen - High-skilled
Total

Education and Citizenship Status (concluded)









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